

Reachability Queries

Time limit: 1.00 s

Memory limit: 512 MB

A directed graph consists of n nodes and m edges. The edges are numbered $1, 2, \dots, n$.

Your task is to answer q queries of the form "can you reach node b from node a ?"

Input

The first input line has three integers n , m and q : the number of nodes, edges and queries.

Then there are m lines describing the edges. Each line has two distinct integers a and b : there is an edge from node a to node b .

Finally there are q lines describing the queries. Each line consists of two integers a and b : "can you reach node b from node a ?"

Output

Print the answer for each query: either "YES" or "NO".

Constraints

- $1 \leq n \leq 5 \cdot 10^4$
- $1 \leq m, q \leq 10^5$

Example

Input	Output
4 4 3 1 2 2 3 3 1 4 3 1 3 1 4 4 1	YES NO YES